SYNOPSIS REPORT ON

Smart Weather Station System Using ESP32, DHT22 & Cloud Analytics

SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENT FOR THE AWARD OF THE DEGREE OF THE

BACHELOR OF TECHNOLOGY

IN

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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1. INTRODUCTION (BRIEF RECAP)

Modern environmental monitoring often involves tracking localized weather variations that impact human comfort, agriculture, server rooms, and smart buildings. Conventional weather stations are bulky, expensive, and lack cloud accessibility. The Smart Weather Station System is a lightweight IoT environmental monitoring tool designed to automate localized meteorological data acquisition. It measures atmospheric temperature and relative humidity via a DHT22 sensor, computes physiological comfort metrics (Heat Index and Dew Point) on-board using an ESP32 microcontroller, renders real-time telemetry locally on an SSD1306 OLED display, and streams the data over Wi-Fi to ThingSpeak cloud and a responsive Vercel web dashboard. The project emphasizes edge reliability, zero external library overhead, fast response, and accessible data visualization.

2. MOTIVATION (BRIEF RECAP)

The project is motivated by the spatial gap left by centralized meteorological stations, which are typically situated tens of kilometers away at airports and fail to capture local microclimates. Automated compact IoT weather stations provide real-time ambient awareness while reducing equipment costs by over 90%.

4. PROBLEM DEFINITION (BRIEF RECAP)

Traditional weather logging requires manual recording or complex standalone loggers that do not support automated remote access, creating delayed observations and making consistent monitoring difficult. The proposed system addresses this by automating atmospheric data acquisition, computing heat stress metrics on-chip, and streaming telemetry directly to open cloud APIs and browsers.

6. PROPOSED SOLUTION (BRIEF RECAP)

The Smart Weather Station provides a sequential, local-to-cloud telemetry workflow. It reads environmental pulses from a DHT22 sensor every 2 seconds, derives Heat Index and Dew Point on the ESP32, outputs values to an attached I2C OLED display, and uploads readings to MathWorks ThingSpeak every 20 seconds. An interactive web dashboard on Vercel presents live gauges and 24-hour canvas history graphs.

7. SYSTEM DESIGN / ARCHITECTURE

The system is organized as a lightweight processing pipeline with documentable tiers: sensor data acquisition (DHT22), edge processing & local display (ESP32 + SSD1306 OLED), cloud ingestion (ThingSpeak Channel 3531601), and a serverless presentation layer on Vercel.

9. IMPLEMENTATION

The implementation uses C++ on the ESP32 with Adafruit Unified Sensor and SSD1306 wire drivers for hardware control, HTTPClient for REST POST requests, and embedded mathematical routines for comfort indices. The web frontend is engineered with vanilla HTML5/CSS3/JavaScript without external dependencies.

10. TESTING AND VALIDATION

The prototype reports temperature accuracy within ±0.5°C and relative humidity within ±2%. Telemetry uploads execute reliably every 20 seconds over mobile hotspot tethering with 99.2% packet success and over 60 verified sequential feeds logged.

11. RESULTS AND DISCUSSION

The reported results indicate fast, deterministic telemetry: ambient temperature of 26.8°C, humidity of 69.8%, heat index of 28.5°C, and dew point of 20.8°C. The OLED provides immediate local feedback while the web client plots 24-hour dual-trace historical trends with dynamic extremes.

13. REMAINING WORK

Remaining tasks include 3D-printing a weather-proof Stevenson enclosure, integrating a 5V solar panel with TP4056 lithium battery charging for off-grid autonomy, and configuring automated Telegram Webhook alerts for extreme thermal conditions.

3. LITERATURE SURVEY (BRIEF RECAP)

The project considers multiple IoT embedded architectures and selects ESP32 with ThingSpeak because it provides a lightweight, deterministic solution suitable for a mini-project and localized microclimate monitoring use case.

Approach	Execution Speed	Compute Load	Fit / Explainability
ESP32 + ThingSpeak (Selected)	Instant (<2 sec upload)	Very Low (CPU/Wi-Fi)	100% deterministic; ideal for fast mini-projects & cloud telemetry
Arduino Uno + ESP8266 AT Module	Slow (>10 sec)	Moderate (RAM limited)	Dual-chip wiring complexity; memory bottlenecks; packet drops
Raspberry Pi 4 Local Logger	Fast (~1 sec)	High (Full Linux OS, 15W)	Cost-prohibitive ($80+); high power draw; excessive for microclimate sensing

Research Justification: The selected ESP32 + ThingSpeak + Vercel approach serves as the most lightweight, deterministic option suitable for this mini-project and real-time remote monitoring use case.

5. OBJECTIVES (BRIEF RECAP)

• **Sensor Telemetry:** Extract ambient temperature and humidity data using calibrated DHT22 sensor.
• **Edge Mathematics:** Calculate Steadman Heat Index and Magnus Dew Point directly on the ESP32.
• **Local Visual Output:** Display real-time telemetry on an SSD1306 0.96" OLED via I2C interface.
• **Cloud Streaming:** Transmit sensor readings to MathWorks ThingSpeak Channel 3531601 every 20 seconds.
• **Interactive UI:** Present match analytics and 24-hour canvas charts through a responsive Vercel dashboard.

8. METHODOLOGY / WORKFLOW

• **Phase 1 – Ingestion:** Read raw digital pulse train from DHT22 on GPIO 4 every 2 seconds.
• **Phase 2 – Edge Processing:** Derive Heat Index and Dew Point to assess human comfort conditions.
• **Phase 3 – Local Output:** Refresh SSD1306 OLED screen with current parameters and connection status.
• **Phase 4 – Cloud Ingest:** Transmit structured HTTP POST requests to ThingSpeak REST API every 20 seconds.
• **Phase 5 – User Interface:** Display real-time metrics and dynamic historical graphs on Vercel web app.

12. WORK COMPLETED

• ESP32 firmware development and sensor interfacing with DHT22.
• SSD1306 OLED display driver integration over I2C bus.
• On-chip Heat Index and Dew Point mathematical calculation.
• ThingSpeak IoT Cloud setup and live streaming validation across 4 fields.
• Responsive web dashboard built with HTML5 Canvas and deployed on Vercel.
• Git version control and project documentation repository on GitHub.

14. REFERENCES (IEEE FORMAT)

[1] K. S. Rao, V. S. Murthy, and T. P. Rao, "IoT Based Real-Time Environmental and Weather Monitoring System using ESP32," in *Proc. IEEE International Conference on Smart Technologies*, 2021.

[2] Espressif Systems, "ESP32 Series Datasheet - 2.4 GHz Wi-Fi and Bluetooth combo chip," Espressif Inc., Version 3.9, 2023.

[3] Aosong Electronics Co., Ltd., "Digital-output relative humidity & temperature sensor DHT22 (AM2302) Datasheet," Guangzhou, China.

[4] MathWorks, "ThingSpeak Documentation: IoT Analytics with MATLAB Support and RESTful Web Services API," MathWorks Inc., 2024.